RahulAloth / inference-engineering-roadmap

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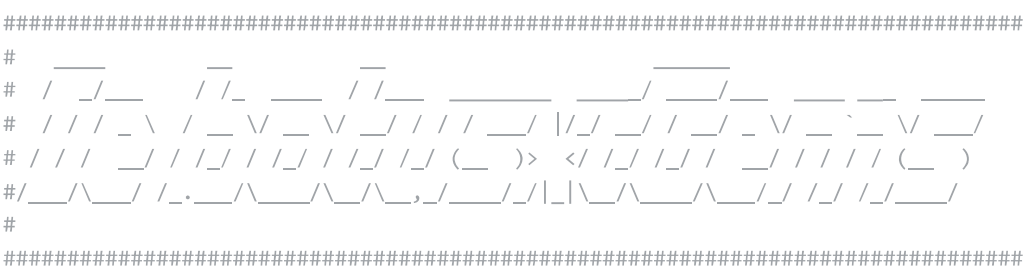
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market_study.md	Create market_study.md	6 months ago



Inference Engineering Roadmap

A complete, end-to-end roadmap for becoming a **production-ready Inference Engineer**, covering ONNX, TensorRT, CUDA, C++, GPU pipelines, and real-time deployment.

🏷️ Badges



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🔍 Overview

This roadmap is designed for engineers transitioning into **AI inference, GPU acceleration, robotics perception, and real-time deployment**.

It focuses on:

- ONNX model export
- TensorRT optimization
- CUDA-based pre/post-processing
- Real-time inference pipelines
- Edge deployment (Jetson/Orin)
- ROS2 integration

You will build **production-grade pipelines**, not just toy ML scripts.

🚀 Roadmap Summary

Week	Focus Area	Output
1	Model literacy + ONNX export	Basic model export repo
2	ONNX Runtime (CPU/GPU)	C++ inference repo

Week	Focus Area	Output
3	TensorRT engine building	FP32/FP16/INT8 comparison
4	GPU pre/post-processing	CUDA pipeline repo
5	Real-time inference pipeline	<20ms end-to-end pipeline
6	Edge deployment (Jetson)	Optimized embedded pipeline
7	ROS2 integration	GPU inference ROS2 node
8	Final project	Full-stack inference system

📅 Week-by-Week Plan

Week 1 — Model Literacy + ONNX Export

- Understand CNNs, segmentation models, transformers
- Export PyTorch → ONNX
- Visualize ONNX graphs

Output: `model_export_basics/`

Week 2 — ONNX Runtime (C++/GPU)

- Load ONNX models in C++
- Bind inputs/outputs
- Measure CPU vs GPU latency

Output: `onnx_runtime_inference/`

Week 3 — TensorRT Fundamentals

- Build TensorRT engines
- FP32 → FP16 → INT8
- Calibration + optimization profiles

Output: `tensorrt_engine_optimization/`

Week 4 — GPU Pre/Post-Processing

- CUDA kernels for resize, normalize, NMS
- Zero-copy GPU pipelines
- Tensor layout conversions

Output: `gpu_prepost_pipeline/`

Week 5 — Real-Time Inference Pipeline

- Async execution
- CUDA streams
- Nsight profiling
- Achieve <20ms latency

Output: `realtime_inference_pipeline/`

Week 6 — Edge Deployment (Jetson Orin/Xavier)

- JetPack
- TensorRT on Jetson
- Power/thermal optimization

Output: `jetson_edge_deployment/`

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- ROS2 nodes
- NITROS zero-copy
- Publish detections at 30–60 FPS

Output: `ros2_gpu_inference_node/`

Week 8 — Final Portfolio Project

Choose one:

- YOLOv8/YOLOv10 detection pipeline
- UNet segmentation
- Multi-camera inference
- Industrial defect detection
- Tracking pipeline (DeepSORT + YOLO)

Output: `fullstack_inference_system/`

📁 Project Folder Structure

```
inference-roadmap/
```

```
├── week01_model_export/
├── week02_onnx_runtime/
├── week03_tensorrt_engines/
├── week04_gpu_prepost/
├── week05_realtime_pipeline/
├── week06_jetson_deployment/
├── week07_ros2_integration/
│
└── final_project/
    ├── src/
    ├── include/
    ├── models/
    ├── tensorrt_engines/
    ├── launch/
    ├── scripts/
    └── README.md
```

🛠️ Recommended Tools

- **PyTorch** (for model export only)
- **ONNX Runtime**
- **TensorRT**
- **CUDA Toolkit**
- **Nsight Systems / Nsight Compute**
- **ROS2 Humble / Iron**
- **JetPack SDK**

📘 Learning Resources

- **NVIDIA TensorRT Developer Guide**
- **ONNX Runtime C++ API**
- **CUDA Programming Guide**
- **ROS2 Tutorials**
- **Jetson Developer Zone**

🏁 Final Portfolio Project

Your final project should demonstrate:

- ONNX export
- TensorRT optimization
- GPU pre/post-processing
- Real-time inference
- ROS2 integration
- Clean C++ architecture

- Latency profiling

This becomes your **flagship GitHub project** for recruiters and hiring managers.

Releases

No releases published

Packages

No packages published

Contributors 1



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